Lesson plan · for the teacher

Who may see what

Data engineering · Level 4 · Lesson 3

Time
40 min

Materials
Fact cards about one made-up pupil; A list of the school's reports and who reads them; Optional: a phone or computer for the SQL step

Learning goals

- Give each role only the data it needs.
- Share summaries, and hide groups under five.
- See how harmless facts can identify a child.

Activity: Guess who

1. Warm-up (5 min): Ask who can see pupils' marks today, and who should.
2. Teach (8 min): Draw an access table on the board. Explain least privilege, summaries and the small-group rule, and why numbers instead of names isn't enough.
3. Play (10 min): Read one fact at a time about a made-up pupil. After each fact, pupils guess who it could be. Count how many facts it took.
4. Decide (12 min): Groups fill in an access table for five records and five roles, and compare their tables.
5. Wrap-up (5 min): Agree one thing the school should never send outside, even without names. Then give the exit ticket.

Make the "Guess who" pupil entirely made up, and choose facts that don't describe a real pupil in your school. Keep the discussion of special needs general.

Discussion

- After how many facts could we tell who she was?
- Why hide groups of fewer than five?
- What should we ask before data leaves the school?

Exit ticket

- Write one record, the roles that may see it, and whether each sees it in full or as a summary.

Try it at school tomorrow
Play "Guess who" with a made-up pupil: give one fact at a time (her class, her age, her bus stop, her sports team). After how many facts can the class tell who she is?

Practice questions · for you

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1 **Every name was replaced with a number before a file was shared. Is the data anonymous now?**

- ◯ No: it's pseudonymised; the list or a few facts can identify a child
- ◯ Only if the file is a CSV
- ◯ Yes: without names, nobody can tell

2 **Match each role to what it should see.**

A class teacher	• • • •	Their own child's marks
A parent	• • • •	Her own class's marks and attendance
The school nurse	• • • •	Summaries, with small groups hidden
The district office	• • • •	Health notes, not marks

3 **The district's report: each class's number of pupils and average mark (to 1 decimal place), hiding any class with fewer than 5 pupils.**

```
SELECT class, COUNT(*), ROUND(AVG(mark), 1) FROM marks
GROUP BY class;
```

Write your program:

4 **A report query returns the row: 6B | hearing | 1 | 7.0. What's the problem?**

```
SELECT class, need, COUNT(*), AVG(mark)
FROM pupils
GROUP BY class, need;
```

- ◯ It points at one child's need and mark
- ◯ AVG gives the wrong answer for one row
- ◯ Nothing: there are no names in it

5 **A table can point at one child even when each of its columns on its own is harmless.**

- ◯ True
- ◯ False

✎ **Draw your school's access table in your head: which record does someone see today that their role doesn't need?**

Check your answers

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1 **No: it's pseudonymised; the list or a few facts can identify a child**

That's pseudonymisation, not anonymisation.

2 **A class teacher — Her own class's marks and attendance**
A parent — Their own child's marks
The school nurse — Health notes, not marks
The district office — Summaries, with small groups hidden

Least privilege: each role gets what its job needs.

3

```
SELECT class, COUNT(*), ROUND(AVG(mark), 1) FROM marks
GROUP BY class
HAVING COUNT(*) >= 5;
```

GROUP BY class makes the summaries; HAVING COUNT(*) >= 5 drops 6C, whose three pupils could be picked out.

4 **It points at one child's need and mark**

A group of one is a person. Everyone who knows 6B knows which child has a hearing need, so the row reveals that child's mark too.

5 **True**

Harmless facts add up. A class, an age and a bus stop are each harmless, but together they can point at one child.